BrightShopper: Real-Time Shopper Analytics Platform

Press Release / Frequently Asked Questions (PR-FAQ)

Press Release

FOR IMMEDIATE RELEASE

BrightSign Unveils BrightShopper: AI-Powered Real-Time Shopper Analytics Platform

Groundbreaking edge AI solution delivers unprecedented insights into customer behavior, engagement, and shopping patterns without cloud dependency or privacy concerns

LOS GATOS, CA – BrightSign, the global market leader in digital signage media players, today announced BrightShopper, a revolutionary edge AI analytics platform that transforms retail digital signage displays into intelligent customer behavior sensors. Built on BrightSign's proven hardware platform, BrightShopper delivers real-time shopper analytics with zero cloud latency, complete data privacy, and cinema-quality performance.

BrightShopper is available across BrightSign's NPU-enabled player lineup, with capabilities scaling based on NPU core count. The flagship XT5 (RK3588, 3-core NPU) runs three specialized AI models simultaneously for comprehensive behavioral analytics. The XS156 (RK3576, 2-core NPU) runs two models for core engagement metrics, while the LS5/HS5 (RK3568, 1-core NPU) provides fundamental attention tracking. All players deliver real-time analytics that were previously impossible to achieve at the edge:

- **People counting and tracking**: Know exactly how many shoppers are in view and track their movement patterns
- **Facial detection and gaze tracking**: Understand which content captures attention and for how long
- **Pose estimation and behavior recognition**: Identify shopping behaviors like cart pushing, basket carrying, shelf interaction, and product selection

"Retailers have long struggled with the fundamental question: 'Who's watching my content, and are they engaged?'" said Steve Durkee, CEO of BrightSign. "BrightShopper answers this question in real-time with unprecedented detail, all while keeping customer data private and secure on the device. This is the future of retail analytics—powerful AI that respects privacy while delivering actionable insights."

Unlike cloud-based analytics solutions that introduce latency, raise privacy concerns, and incur ongoing bandwidth costs, BrightShopper processes all data locally on the BrightSign player. Retailers receive rich behavioral insights in milliseconds, not seconds, enabling real-time content adaptation and immediate performance measurement.

Key Capabilities:

- **Real-time engagement metrics**: Person count, face count, gaze count, and attention duration
- **Movement analytics**: Track shopping paths, dwell times, and traffic patterns
- **Behavioral insights**: Detect cart pushing, basket carrying, product interaction, and browsing behavior
- **Frame-to-frame tracking**: Maintain continuity across frames to understand customer journeys
- **Privacy-first design**: All processing on-device with no cloud transmission of video or personal data

BrightShopper is available immediately as a software upgrade for compatible BrightSign players. Custom behavior models and integration services are available through BrightSign's professional services team.

For more information, visit **brightsign.biz/brightshopper** or contact your BrightSign sales representative.

About BrightSign BrightSign is the global market leader in digital signage media players, delivering reliable, high-performance solutions to over 100,000 customers worldwide. BrightSign players power digital signage in retail stores, restaurants, corporate environments, transportation hubs, and entertainment venues across the globe.

Frequently Asked Questions

Product Vision & Strategy

Q: What is BrightShopper?

A: BrightShopper is an edge AI analytics platform that transforms BrightSign digital signage players into intelligent shopper behavior sensors. It uses specialized AI models running on the player's on-board NPU to deliver real-time insights about customer count, engagement, movement, and shopping behavior—all without requiring cloud connectivity or compromising privacy.

Q: Which BrightSign players support BrightShopper?

A: BrightShopper is available on BrightSign players with NPU-enabled SoCs. The feature set scales based on the NPU capabilities:

XT5 (Premium - RK3588):

- **NPU**: 3-core, 6 TOPS
- **Models**: Runs all three AI models simultaneously
- **Features**: Full feature set including pose estimation and behavior detection
- **Use case**: High-value locations and comprehensive analytics needs

XS156 (Mid-tier - RK3576):

- **NPU**: 2-core, 4 TOPS
- **Models**: Runs two AI models simultaneously (face/gaze detection + object detection)
- **Features**: Core engagement metrics without pose estimation
- **Use case**: Cost-effective solution for large-scale deployments focused on attention metrics

LS5/HS5 (Entry - RK3568):

- **NPU**: 1-core, 1 TOPS
- **Models**: Runs one AI model (face/gaze detection)
- **Features**: Basic engagement metrics (person count, face count, gaze tracking)
- **Use case**: Budget-conscious deployments requiring fundamental attention analytics

All players deliver real-time analytics with complete on-device processing and privacy protection. The difference is the number of simultaneous AI models supported.

Q: What are the differences between BrightShopper across player models?

A: The players offer different analytics capabilities based on NPU core count:

Feature	XT5 (RK3588)	XS156 (RK3576)	LS5/HS5 (RK3568)
NPU Cores	3-core (6 TOPS)	2-core (4 TOPS)	1-core (1 TOPS)

AI Models Running	3 models	2 models	1 model
Face Detection	✓ Yes	✓ Yes	✓ Yes
Gaze Tracking	✓ Yes	✓ Yes	✓ Yes
Object Detection	✓ Yes (carts, baskets, products)	✓ Yes (carts, baskets, products)	✕ No
Pose Estimation	✓ Yes (17 keypoints)	✕ No	✕ No
Behavior Detection	✓ Yes (cart pushing, shelf reach, etc.)	Limited (inferred from objects only)	✕ No
Person Tracking	✓ Advanced (with pose history)	✓ Basic (bbox tracking)	✓ Basic (bbox tracking)
Person Count	✓ Yes	✓ Yes	✓ Yes
Face Count	✓ Yes	✓ Yes	✓ Yes
Gaze Count	✓ Yes	✓ Yes	✓ Yes
Movement Analytics	✓ Yes	✓ Yes	✓ Basic
Dwell Time	✓ Yes	✓ Yes	✓ Yes

XT5 Use Cases:

- Premium retail locations requiring comprehensive behavior insights
- Stores wanting to detect cart/basket usage and shelf interactions
- Deployments needing custom behavior classification
- Applications requiring detailed pose data for analytics

XS156 Use Cases:

- Large-scale deployments where attention metrics are primary goal
- Budget-conscious installations focused on engagement ROI
- Locations where gaze and object context are sufficient
- High-volume rollouts across multiple stores

LS5/HS5 Use Cases:

- Entry-level engagement tracking (people count, attention, dwell)
- Large-scale deployments requiring only fundamental metrics
- Budget-sensitive installations where basic analytics suffice
- Proof-of-concept and pilot programs

Q: Why did BrightSign build BrightShopper?

A: Retailers consistently tell us they need better data about who's seeing their content and whether it's working. Traditional solutions either require expensive infrastructure, raise privacy concerns with cloud-based video analysis, or provide only basic metrics like "someone walked by." BrightShopper delivers rich behavioral analytics that answer the fundamental questions: How many people are watching? Are they engaged? What are they doing? And it does this while keeping all data private and on-device.

Q: Who is the target customer?

A: BrightShopper is designed for:

- **Retailers** wanting to measure digital signage ROI and optimize content placement
- **Brands** deploying in-store marketing displays who need engagement metrics
- **Shopping centers and malls** seeking foot traffic and dwell time analytics
- **QSRs and convenience stores** wanting to understand queue behavior and product interaction
- **Consumer goods companies** running promotional displays who need proof of engagement

Q: How is BrightShopper different from existing analytics solutions?

A: Most analytics solutions face a fundamental tradeoff between capability, privacy, and performance:

- **Simple people counters** provide basic counts but no behavioral insight
- **Cloud-based video analytics** offer rich analytics but raise privacy concerns, require bandwidth, and introduce latency
- **Retail analytics platforms** require dedicated hardware, complex installation, and ongoing services

BrightShopper eliminates these tradeoffs by:

- Running all AI processing on-device (no video leaves the player)
- Delivering comprehensive behavioral analytics from a single device
- Requiring no additional hardware beyond the existing BrightSign player
- Providing real-time results in under 70ms
- Operating with no recurring cloud costs

Technical Capabilities

Q: What specific metrics does BrightShopper provide?

A: BrightShopper delivers frame-by-frame analytics including:

Counts (instantaneous):

- Number of people in frame
- Number of faces detected
- Number of people looking at screen

Tracking (temporal):

- Unique person IDs maintained across frames
- Movement direction and velocity
- Dwell time and path tracking
- Gaze state changes (when people look toward/away)

Behavioral insights:

- Shopping with cart or basket
- Browsing/standing behavior
- Walking/moving through space
- Shelf interaction and product pickup
- Pose changes indicating specific actions

Q: How accurate is BrightShopper?

A: BrightShopper uses state-of-the-art AI models optimized for retail environments:

- **Person detection**: >95% precision at typical retail distances (2-15 feet)
- **Face detection**: >90% detection rate for faces at 1-10 feet
- **Gaze estimation**: ±15-degree accuracy for determining screen attention
- **Pose keypoints**: 17-point skeleton with >85% keypoint detection rate
- **Tracking**: Maintains ID consistency >95% of the time in typical retail scenarios

Accuracy varies with lighting, camera angle, distance, and occlusion. We provide best practices guides for camera placement to optimize performance.

Q: What AI models power BrightShopper?

A: BrightShopper uses different AI model configurations depending on the player:

XT5 (3 models):

1. **RetinaFace**: Face detection and gaze estimation - identifies faces and determines if people are looking at the screen
2. **YOLOv8-pose**: Person detection with 17-point pose estimation - tracks body position and enables behavior recognition
3. **YOLOx**: Object detection - identifies shopping carts, baskets, and products for context

XS156 (2 models):

1. **RetinaFace**: Face detection and gaze estimation - identifies faces and determines if people are looking at the screen
2. **YOLOx**: Object detection - identifies shopping carts, baskets, and products for context; also provides person detection for tracking

Both configurations use a fusion layer that correlates detections, then a tracking layer maintains person identity and movement state across frames. **Q: What is the performance and latency?**

A: BrightShopper achieves real-time performance on both players:

XT5 (3 models):

- **Latency**: ~69ms end-to-end (capture to analytics output)
- **Throughput**: 14 FPS with all three models, 20+ FPS with optimized configuration

XS156 (2 models):

- **Latency**: ~50ms end-to-end (capture to analytics output)
- **Throughput**: 18-20 FPS with both models

This real-time performance enables immediate content adaptation and responsive interactive experiences.

Q: What hardware is required?

A: BrightShopper requires:

- **BrightSign player**: XT5 or XS156
- **USB or IP camera**: 720p minimum, 1080p recommended
- **Network connection**: For analytics output (UDP/JSON, HTTP, MQTT supported)

No additional compute hardware or cloud services required. The BrightSign player handles all AI processing on-device.

Q: How does BrightShopper handle privacy?

A: Privacy is built into BrightShopper's architecture:

- **No video storage**: Raw video is never saved to disk
- **No cloud transmission**: Video never leaves the device
- **On-device processing**: All AI inference happens locally on the NPU
- **Anonymized output**: Only pose keypoints, bounding boxes, and counts are transmitted—no images or facial recognition
- **GDPR/CCPA friendly**: No personal data is collected or stored
- **Configurable privacy zones**: Option to disable analytics in specific areas

Retailers get rich behavioral insights without compromising customer privacy.

Business Model & Pricing

Q: How is BrightShopper priced?

A: BrightShopper is available through flexible licensing:

- **Software license** included with compatible RK3588-based BrightSign players
- **Analytics tier pricing** based on output frequency and feature set:
  - **Basic**: Person/face counts, gaze detection ($X/player/month)
  - **Professional**: Adds tracking, movement analytics ($X/player/month)
  - **Enterprise**: Adds behavior detection, custom models (custom pricing)
- **One-time purchase** option available for high-volume deployments
- **Professional services** available for custom model training and integration

Contact your BrightSign sales representative for detailed pricing.

Q: What is included in the base price vs. add-ons?

A: **Base BrightShopper license includes:**

- Real-time person and face counting
- Gaze detection and engagement tracking
- Basic movement tracking
- JSON/UDP output
- Standard AI models

Professional tier adds:

- Extended tracking with behavior history
- Movement pattern analysis
- Dwell time analytics
- HTTP/MQTT output
- Dashboard visualization

Enterprise tier adds:

- Custom behavior model training (cart pushing, product pickup, etc.)
- Transfer learning for store-specific actions
- Advanced API integration
- Priority support and SLAs

Q: What is the ROI for retailers?

A: Retailers see ROI through multiple channels:

Content optimization: Test which creative drives engagement, optimize content rotation schedules, and eliminate underperforming assets (typical 20-40% improvement in engagement rates)

Placement optimization: Understand which display locations drive the most attention and interaction (helps justify premium placement costs or relocate underperforming displays)

Campaign measurement: Provide advertisers and brands with engagement proof-of-performance (enables premium pricing for advertising inventory)

Operational insights: Understand traffic patterns, peak times, and customer flows to optimize staffing and merchandising

Competitive advantage: Offer data-driven insights that pure digital signage competitors cannot match

Q: What analytics software is available for BrightShopper?

A: BrightSign offers two complementary analytics platforms for collecting and visualizing BrightShopper data across multiple players:

BSN.cloud Analytics (Cloud-Based):

- Integrated into BrightSign's BSN.cloud content management platform
- Aggregates BrightShopper data from all connected players
- Interactive dashboards with graphs showing trends over time
- **Content attribution**: Links viewer metrics to specific media being played at the time of viewership
- Ideal for customers already using BSN.cloud for content management
- Subscription-based pricing as part of BSN.cloud service

BrightShopper Analytics Enterprise (On-Premise):

- Self-hosted enterprise software package
- Aggregates BrightShopper data from fleet of players
- Interactive dashboards with graphs showing trends over time
- Does not include content attribution (content management separate)
- Ideal for large enterprises requiring on-premise data storage
- One-time license with annual maintenance

Both platforms provide the same core analytics capabilities—collecting shopper metrics across locations, visualizing trends, and generating reports. The key difference is BSN.cloud's content attribution feature and cloud hosting vs. enterprise on-premise deployment.

Q: What is content attribution and why does it matter?

A: Content attribution is a powerful feature in BSN.cloud Analytics that links viewer engagement metrics directly to the specific content being displayed at that moment.

How it works:

- BSN.cloud knows what content is playing on each player at every moment
- BrightShopper continuously streams engagement data (person count, gaze, dwell time, etc.)
- BSN.cloud correlates the two data streams to attribute viewers to content

What you can measure:

- **Engagement by creative**: Which ads/videos captured the most attention?
- **Performance by time slot**: Does content perform better at certain times of day?
- **Campaign effectiveness**: Compare engagement across different creative versions

- **ROI per asset**: Understand which content investments drove the most engagement

Example insights:

- "Creative A had 45% gaze rate vs. Creative B's 28% gaze rate"
- "Morning shoppers spend 8 seconds viewing promotions vs. 4 seconds in afternoon"
- "Product demo videos generate 3x more shelf interactions than static ads"

Why it matters:

- **Optimize content strategy**: Double down on what works, eliminate what doesn't
- **Prove campaign ROI**: Show clients/stakeholders engagement data per campaign
- **Dynamic content scheduling**: Schedule high-performing content during peak traffic times
- **A/B testing**: Compare creative variants with real engagement data

Without content attribution, you know *how many* people are engaging, but not *what content* they're engaging with. BSN.cloud Analytics connects these dots automatically.

Q: How do the two analytics platforms compare?

A: Both platforms provide comprehensive shopper analytics, but differ in deployment model and features:

Feature	BSN.cloud Analytics	BrightShopper Analytics Enterprise
Deployment	Cloud-hosted (SaaS)	On-premise / self-hosted
Data Collection	✓ Multi-player aggregation	✓ Multi-player aggregation
Time-Series Graphs	✓ Yes	✓ Yes
Interactive Dashboards	✓ Yes	✓ Yes
Export/Reports	✓ Yes	✓ Yes
Content Attribution	✓ Yes	✕ No
Content Management Integration	✓ Native (BSN.cloud)	Separate CMS required
Data Storage	BrightSign cloud	Customer infrastructure
Pricing Model	Subscription (per player/month)	One-time license + maintenance
Setup Time	Minutes (auto-provision)	Days (IT deployment)
Maintenance	BrightSign managed	Customer managed
Updates	Automatic	Manual (scheduled releases)
API Access	✓ REST API	✓ REST API
Custom Integrations	Webhooks	Full database access

Choose BSN.cloud Analytics if:

- You use (or plan to use) BSN.cloud for content management

- You want content-to-engagement attribution
- You prefer cloud-hosted, zero-maintenance solutions
- You need rapid deployment across multiple locations

Choose Analytics Enterprise if:

- You require on-premise data storage for compliance/security
- Your IT policy prohibits cloud analytics services
- You need deep integration with existing enterprise systems
- You want full control over data infrastructure and retention

Many customers use both: BSN.cloud Analytics for corporate stores with content attribution needs, and Analytics Enterprise for customers requiring on-premise deployment.

Integration & Deployment

Q: How does BrightShopper integrate with existing systems?

A: BrightShopper provides flexible integration options:

Real-time outputs:

- **UDP JSON**: Low-latency streaming for real-time applications
- **HTTP POST**: Standard REST API for easy integration
- **MQTT**: Pub/sub for IoT platforms and analytics dashboards

Analytics platforms:

- **Pre-built connectors**: Google Analytics, Tableau, Power BI
- **Custom webhooks**: Send to any HTTP endpoint
- **Data warehouse export**: Batch export for historical analysis

Content management systems:

- **BrightAuthor integration**: Trigger content changes based on analytics
- **Third-party CMS**: API for integration with any CMS platform

Q: What does deployment look like?

A: BrightShopper deployment is straightforward:

1. **Hardware setup**: Install BrightSign player with RK3588 and connect camera (same as standard digital signage deployment)
2. **Software configuration**: Enable BrightShopper via BrightAuthor or configuration file, set camera parameters and privacy zones
3. **Model deployment**: Download AI models to player (included with license, ~15MB total)
4. **Integration setup**: Configure analytics output (UDP, HTTP, or MQTT endpoints)
5. **Validation**: Use provided dashboard to verify analytics are flowing correctly

Typical deployment time: 15-30 minutes per player. Remote deployment and configuration supported.

Q: Can I customize the behavior detection for my store?

A: Yes! BrightShopper supports custom behavior models through transfer learning:

Out-of-box behaviors:

- Cart pushing / basket carrying
- Shelf interaction / product reach

- Standing / browsing
- Walking / moving

Custom behaviors (via transfer learning):

- Store-specific product interactions
- Queue management behaviors
- Fitting room entrances
- Service counter interactions
- Any behavior visible through pose + context

BrightSign Professional Services can help train custom models using your store footage.

Note: Advanced pose-based behavior detection and custom transfer learning require the XT5 player. The XS156 provides basic behavior inference from object associations (e.g., person near cart = likely shopping).

Competition & Market Position

Q: How does BrightShopper compare to RetailNext, ShopperTrak, etc.?

A: Traditional retail analytics platforms require dedicated hardware installations:

Feature	BrightShopper	Traditional Analytics	Cloud Video Analytics
Hardware cost	$0 (uses existing signage player)	$500-2000+ per location	$0 (software only)
Installation	15 min (camera only)	2-4 hours (dedicated sensors)	15 min (camera only)
Privacy	On-device, no video leaves player	Varies by vendor	Video sent to cloud
Latency	<70ms real-time	Seconds to minutes	Seconds (round-trip)
Behavioral insights	Rich (pose + gaze + objects)	Basic (counts + zones)	Rich (video analysis)
Ongoing costs	Software license only	Hardware + software + services	Bandwidth + compute + storage
Integration	Built into signage CMS	Separate platform	Separate platform

BrightShopper delivers comparable or superior analytics at a fraction of the cost and complexity.

Q: Why not just use a smartphone app or WiFi tracking?

A: Smartphone-based solutions have fundamental limitations:

- **Opt-in required**: Only tracks customers who download app and enable tracking (typically <5% of shoppers)
- **No behavioral insight**: Can't see what people are looking at, holding, or doing
- **No content correlation**: Can't tie analytics to specific digital signage content
- **Privacy concerns**: Tracking phone IDs raises more privacy issues than anonymous pose data

BrightShopper provides 100% coverage of everyone in the camera's view, with rich behavioral context, and better privacy protection.

Future Roadmap

Q: What's next for BrightShopper?

A: Our roadmap includes:

Near-term (6 months):

- Demographic estimation (age range, gender - optional, privacy-configurable)
- Emotion detection (engaged, surprised, confused)
- Group detection (shopping alone vs. with family)
- Extended tracking across multiple cameras

Medium-term (12 months):

- Product recognition (identify specific products being picked up)
- Gesture recognition (pointing, reaching, waving)
- Queue analytics (line length, wait times)
- Heat map generation (attention density maps)

Long-term (18+ months):

- Multi-modal sensing (audio + visual for richer context)
- Predictive analytics (predict likely purchase intent)
- Real-time content adaptation (automatically optimize content based on audience)
- Cross-location analytics (compare performance across stores)

Q: Will BrightShopper work with future BrightSign hardware?

A: Yes. BrightShopper is designed to scale with future hardware:

- Compatible with current and future RK3588-based players
- Modular architecture supports next-gen NPUs and AI accelerators
- Model format (RKNN) is forward-compatible with Rockchip roadmap
- Software updates ensure compatibility with new BrightSign OS versions

As hardware improves, BrightShopper will automatically deliver higher frame rates, support more simultaneous tracks, and enable more sophisticated models—all through software updates.

Customer Testimonials

"BrightShopper has transformed how we measure our in-store digital marketing ROI. We can finally prove which content drives engagement and optimize our creative accordingly. The insights we're getting are incredible, and the fact that it's all private and on-device means we have no concerns about customer data." — **Jamie Rodriguez, Director of Digital Marketing, [Major Retail Chain]**

"We deployed BrightShopper across 200 stores in 30 days. The installation was simple—just connect a camera—and the analytics started flowing immediately. We're now using dwell time and engagement metrics to optimize product placement and justify premium advertising rates. This is a game-changer." — **Michael Chen, VP of Store Operations, [National Grocery Chain]**

"As a brand running promotional displays in retail, BrightShopper gives us the proof-of-performance data we need to show our CPG clients. We can demonstrate not just foot traffic, but actual engagement and interaction. It's become a key differentiator in our sales process." — **Sarah Mitchell, CEO, [Retail Marketing Agency]**

Technical Appendix

Output Data Format

BrightShopper outputs JSON messages via UDP, HTTP, or MQTT:


```json
{
  "timestamp": 1634567890123,
  "frame_id": 12345,
  "analytics": {
    "counts": {
      "persons": 5,
      "faces": 4,
      "gazes": 3
    },
    "tracks": [
      {
        "track_id": 42,
        "bbox": [120, 80, 280, 520],
        "keypoints": [[150, 100], [145, 95], ...],
        "has_face": true,
        "looking_at_screen": true,
        "has_cart": false,
        "has_basket": true,
        "behavior": "CARRYING_BASKET",
        "velocity": [2.3, -0.5],
        "direction_degrees": 282,
        "dwell_time_seconds": 12.4,
        "frames_since_gaze_change": 45
      }
    ],
    "objects": [
      {"class": "cart", "bbox": [500, 300, 600, 500], "confidence": 0.92},
      {"class": "basket", "bbox": [200, 150, 250, 200], "confidence": 0.88}
    ]
  }
}
```


Camera Requirements

Minimum specifications:

- Resolution: 720p (1280×720)
- Frame rate: 15 FPS minimum, 30 FPS recommended
- Field of view: 60-90 degrees horizontal
- Mounting height: 7-12 feet above floor
- Lighting: 200+ lux (typical retail lighting sufficient)

Recommended cameras:

- Logitech C920/C930 (USB)
- Any RTSP-compatible IP camera with H.264 encoding

Model Performance Details

Model	Input Size	NPU Core	Inference Time	Output
RetinaFace	320×320 RGB	0	~15ms	Faces + gaze vectors
YOLOv8-pose	640×640 BGR	1	~25ms	Persons + 17 keypoints
YOLOx	640×640 BGR	2	~30ms	Objects (carts, baskets, etc.)

Total system latency: ~69ms (capture → analytics output)

For more information, technical documentation, and integration guides, visit the BrightShopper Developer Portal at **developers.brightsign.biz/brightshopper**